

Ledger Engine Documentation

Overview

The Ledger Engine maintains three independent ledgers:

1. PSP Ledger
2. NPCI Ledger
3. Bank Ledger

- Each ledger is implemented as a separate Spring Boot microservice.
- Communication between services is asynchronous using Apache Kafka.
- Data is persisted in PostgreSQL.

1. PSP Ledger Service

Purpose

Stores transaction initiation details generated by the PSP Switch when a user starts a UPI payment.

Producer Service

PSP Switch

When a customer initiates a payment:

alice@sbi → bob@hdfc
₹500

the PSP Switch publishes an event.

Kafka Topic

Upi.transactions.initiated

Producer Message

Produced by: PSP Switch

Message:

```
{  
  "txnId": "TXN1001",  
  "payer": "alice@sbi",  
  "payee": "bob@hdfc",  
  "amount": 500  
}
```

Consumer Service

PSP Ledger Service

Consumes:

upi.transactions.initiated

Database Table

psp_ledger

Column	Description
id	Primary Key
txn_id	Transaction Identifier
payer_vpa	Sender VPA
payee_vpa	Receiver VPA
amount	Transaction Amount
status	INITIATED

Creating PSP Ledger Table

```
CREATE TABLE psp_ledger (  
  id SERIAL PRIMARY KEY,  
  txn_id VARCHAR(100) NOT NULL,  
  payer_vpa VARCHAR(100) NOT NULL,  
  payee_vpa VARCHAR(100) NOT NULL,  
  amount DOUBLE PRECISION NOT NULL,  
  status VARCHAR(50) NOT NULL  
);
```

2. NPCI Ledger Service

Purpose

- Stores routing and verification information processed by NPCI.
- NPCI acts as the central switch between payer bank and payee bank.

Producer Service

NPCI Router

After receiving the request from PSP Switch and validating it, NPCI publishes a verification event.

Kafka Topic

upi.npci.verified

Producer Message

Produced by: NPCI Router

Message:

```
{  
  "txnId": "TXN1001",  
  "sourceBank": "SBI",  
  "destinationBank": "HDFC",  
  "amount": 500 }  
}
```

Consumer Service

NPCI Ledger Service

Consumes:

upi.npci.verified

Database Table

npci_ledger

Column	Description
id	Primary Key
txn_id	Transaction Identifier
source_bank	Payer Bank
destination_bank	Payee Bank
amount	Transaction Amount
status	VERIFIED

Creating NPCI Ledger Table

```
CREATE TABLE npci_ledger (  
  id SERIAL PRIMARY KEY,  
  txn_id VARCHAR(100) NOT NULL,  
  source_bank VARCHAR(100) NOT NULL,  
  destination_bank VARCHAR(100) NOT NULL,  
  amount DOUBLE PRECISION NOT NULL,  
  status VARCHAR(50) NOT NULL  
);
```

3. Bank Ledger Service

Purpose

- Stores actual accounting entries after CBS (Core Banking System) processing.
- Bank Ledger follows double-entry accounting.
- Every transaction generates:
 - ☐ 1 Debit Entry
 - ☐ 1 Credit Entry

Debit Flow

Producer Service

CBS Service

After successfully debiting the payer account:

alice@sbi

CBS publishes:

Kafka Topic

upi.cbs.debit.confirm

Producer Message

```
{  
  "txnId": "TXN1001",  
  "account": "alice@sbi",  
  "entryType": "DEBIT",  
  "amount": 500  
}
```

Consumer Service

Bank Ledger Service

Consumes: upi.cbs.debit.confirm

Credit Flow

Producer Service

CBS Service

After successfully crediting:

bob@hdfc

CBS publishes:

Kafka Topic

upi.cbs.credit.confirm

Producer Message

```
{  
  "txnId": "TXN1001",  
  "account": "bob@hdfc",  
  "entryType": "CREDIT",  
  "amount": 500  
}
```

Consumer Service

Bank Ledger Service

Consumes:

upi.cbs.credit.confirm

Database Table

bank_ledger

Column	Description
id	Primary Key
txn_id	Transaction Identifier
account	Account/VPA

entry_type DEBIT or CREDIT
amount Transaction Amount
status SUCCESS

Creating Bank Ledger Table

```
CREATE TABLE bank_ledger (  
  id SERIAL PRIMARY KEY,  
  txn_id VARCHAR(100) NOT NULL,  
  account VARCHAR(100) NOT NULL,  
  entry_type VARCHAR(20) NOT NULL,  
  amount DOUBLE PRECISION NOT NULL,  
  status VARCHAR(50) NOT NULL  
);
```